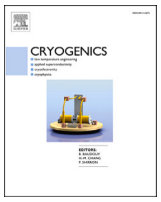
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Two stage pulse tube cryocooler with intermediate heat exchanger for accessing regenerator cooling capacity

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ABSTRACT

Conventional operation of closed-cycle two-stage GM-type Pulse Tube Cryocoolers (PTCs) usually relies on utilizing the cooling power of both the 1st and the 2nd stage. While the 1st stage is required to precool the 2nd stage to reach the lowest accessible temperature below 4K, it usually also provides enough cooling power to cool additional cryostat elements such as radiation shieldings. However, current applications in quantum physics have highlighted the need to additionally access heat sinks with intermediate temperatures and cooling powers, e.g. for cooling of superconducting wires. Here we will demonstrate a cooler configuration, where a third cooling stage is incorporated into the 2nd stage regenerator. This third intermediate cooling stage allows to extract 4-5 W of cooling power at temperatures between 8 K and 9 K for a standard two-stage PTC with a cooling capacity of 1.6 W at 4.2 K. Most importantly, this approach does not reduce the performance of the main stage but the added intermediate regenerator stage instead allows to tap into hidden cooling power of the PTC.

1. Introduction

Ever since their introduction in 1996 [1], closed-cycle two-stage GM-type Pulse Tube Cryocoolers (PTCs) have proven to be most valuable tools for a large variety of cooling applications in cryotechnology [2]. The cooling is facilitated by the principle of cyclic expansion of Helium as working fluid that is compressed in a remote compressor. By connecting two cooling-stages, temperatures as low as 2.1 K can be reached for the 2nd cooling-stage while the 1st stage is typically limited to temperatures of about 40 K.

Nowadays, such PTCs are routinely applied in many different cooling scenarios. One field of application is the cooling of components which are in direct contact with the cooling stage. This approach can, for instance, be used in metrology, where high precision voltage standards need to be operated at cryogenic temperatures [3]. Similarly, detectors for either electromagnetic or particle radiation can directly be cooled, as was e.g. demonstrated for IR-detectors in the field of astronomy [4,5]. This approach of direct cooling can become particularly important for applications where any use of liquid coolants is not acceptable, as was e.g. the case for operating a cryogenic infrared detector in an airplane [4].

In such cases, the cooling strategies rely exclusively on the cooling power of the two stages. While the 1st stage is commonly used to cool thermal radiation shielding and precool wires, the 2nd stage is employed to provide cooling power for the experiment/application. Although this approach is feasible in many cases, it does not fully utilize the overall cooling power of a two-stage PTC.

This can be seen through a common technical solution to enhance the liquefaction capabilities of PTCs. In general, PTCs are becoming increasingly attractive for liquefaction of He^3 and He^4 due to their continuously increasing cooling powers. On one hand, this is relevant to reduce Helium consumption of existing commercial machines, like e.g. NMR-scanners, by re-liquefaction in a closed cycle zero boil off system. On the other hand, the capability to liquify Helium allows to build customized closed-cycle cryostats for specific purposes or more generally cryostats designed to accommodate existing experiments that are built on dipsticks for liquid Helium (LHe) dewars. In such setups, the He gas is then often pre-cooled through thermal contact with the 2nd stage regenerator. To this end, a spiral gas capillary is used as a heat exchanger

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along the full length of the regenerator before final liquefaction is done at the 2nd stage.

While this approach clearly works [6], it is difficult to quantify how much cooling power can actually be extracted from the regenerator and how this influences the main cooling stages [7–10]. More importantly, the cooling power is not extracted at any specific temperature but instead the gas moves along the whole temperature gradient profile of the regenerator.

However, current applications, like e.g. additional cooling of superconducting wires in quantum technology systems, would largely benefit from a more specific approach where a heat sink is located at a specific position and well-defined cooling power can be extracted for a certain temperature. Thereby, the two existing cooling stages would be complemented by a third cooling stage with intermediate capabilities that can, to a certain extent, be tailored to specific requirements.

Consequently, the analysis of such configurations has been the subject of numerous publications, related to both theoretical and experimental aspects of regenerator heating. Particular interest was directed towards the temperature and energy transport profiles across the regenerator which can either be described by analytical or numerical calculations [11,12]. Later on, the results can, for instance, be used to optimize heat load distributions across the regenerator [13]. It can be argued, that such an extraction of additional cooling power increases the overall coefficient of performance (COP) of cryocoolers [14].

From an experimental point of view, applying heat and measuring temperature along the regenerator can be difficult, since the usually employed copper clamps can be problematic and good thermal contact is not necessarily ensured.

Therefore, the large potential for accessing additional cooling power of the regenerator is now demonstrated by cooling data of a novel two stage GM-Type Pulse tube cooler (TransMIT PTD 420RX). This cryocooler provides a 2nd stage cooling power of 1.6 W at 4.2 K and was modified by brazing a Copper stage with optimized thermal contact to the inner working fluid into the 2nd stage regenerator. The new PTD 420RX then retains its original cooling power for the 2*nd* cooling stage while simultaneously providing significant additional cooling power of 4-5 W at temperatures around 9 K.

The position in height of the additional stage is discussed and findings are supported by the comparison of the experimental data with corresponding REGEN 3.3 simulations. The latter also conclude, that the additional copper stage integrated into the regenerator tubing does not significantly modify the overall cooling capacity of the cold head characteristics. Instead, the new regenerator stage elevates the overall cooling power and versatility.

2. Experimental setup

The experiments have been performed using two different PTCs. First a prototype PTC (TransMIT GmbH, Giessen, Germany) was used that was later modified by brazing a Copper stage with optimized thermal contact into the 2nd stage regenerator (then called PTD 420RX). For both coolers, the 1st stage regenerator was filled with an experimental stacked filling of Pb and stainless steel while the 2nd stage regenerators were filled with GOS, $HoCu_2$, Er_3Ni and Pb (see below for further details).

The PTCs were driven by a F-100 Helium compressor (Sumitomo Heavy Industries, Ltd.) filled with an intermediate pressure of 1.7 MPa and operated at the power grid frequency of 50 Hz. The frequency of the pressure oscillation was set to 1.1 Hz and the flexible connection lines from compressor to the rotary valves of the PTCs were 20 m long with an inner diameter of 25 mm. All measurements were performed with the cold part of the PTCs inside a vacuum chamber under sufficiently high vacuum conditions. In each case, the 2nd stage was surrounded by a Copper radiation shield, which was thermally coupled to the cold flange of the 1st stage. In addition, both stages, the radiation shield, 1st stage pulse tube, and the regenerator tubes as well as intermediate stage were wrapped in several layers of aluminized Mylar foil. The temperature T_2 of the cold flange of the 2nd stage and the temperature of the regenerator intermediate stage were measured using Cernox (CX1050-AA, LakeShore) resistance thermometers, while the 1st stage temperature T_1 was recorded using a standard Pt100 (class A) resistance thermometer. Pressure readings were taken from the compressor. Each of the two PTCs was optimized by adjusting the orifice and double inlet valves for optimum cooling power near 4 K on the 2nd stage. After that, the cooling power of all prototypes could be quantified by applying heat loads to both cold stages and the regenerator intermediate stage using electrical resistors.

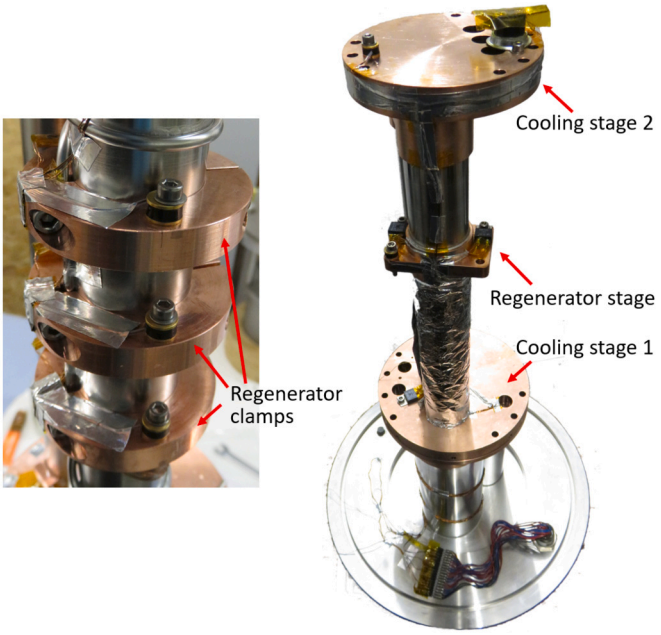


Fig. 1. Regenerator clamps added to the prototype PTC (left) and picture of the soldered regenerator stage of the PTD 420RX (right).

First tests aimed at characterizing the suitable position for the intermediate stage. To this end, 3 loop type clamps that could each be fitted with both an electrical heater and a Cernox resistance thermometer were mounted on the 2nd stage regenerator of the prototype PTC (cf. Fig. 1, left).

The three clamps were distributed along the lower part of the regenerator that is filled with rare earth materials. More specifically, the lowest clamp (i.e. closest to the 2nd stage) was positioned at the transition between GOS and $HoCu_2$, the 2nd clamp in the middle of the $HoCu_2$ layer, and finally the third clamp at the transition between $HoCu_2$ and Er_3Ni.

By applying thermal loads to the 1st stage, the 2nd stage and one of the clamps, a first cursory characterization of the regenerator stages was done. Our results showed that both the lower and upper position were not well suited. Applying heat to the lower position (i.e. GOS/$HoCu_2$ transition) almost immediately affected the 2nd stage temperature, while the upper position (i.e. $HoCu_2$/Er_3Ni transition) was already too close to the 1st stage and did not yield temperatures far below 10 K.

As a consequence, the middle position was chosen for further experiments, since this position provided minimum temperatures around 3 K and additional heating left the 2nd stage largely unaffected.

However, the thermal contact between the stainless steel regenerator tube and the Copper clamp was not ideal and the influence of thermal contact resistances became evident by showing unrealistically high cooling powers, i.e. thermal equilibrium within the clamps was not reached.

Therefore, the thermal contact of the regenerator stage was improved by replacing part of the stainless steel tube along the $HoCu_2$ filling by a brazed-in Cu block with a width of appr. 20 mm to act as

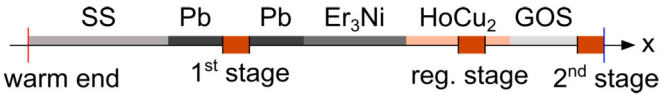


Fig. 2. Schematic representation of the regenerator filling and the positioning of the 3 cooling stages for the PTD 420RX. In this configuration, the warm end is always kept at room temperature.

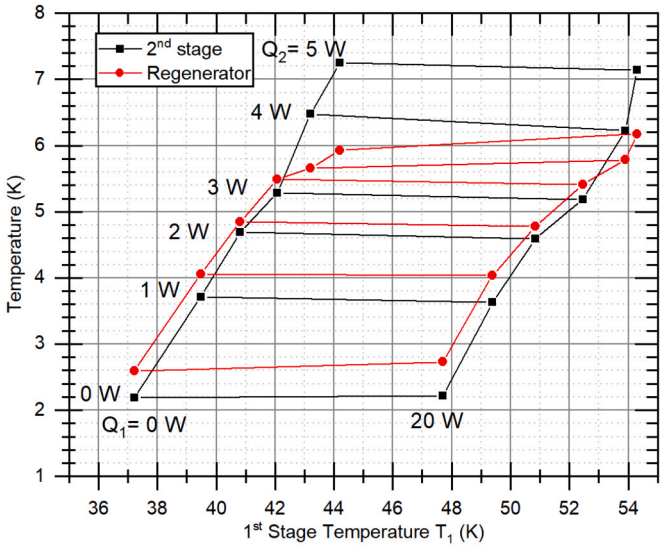


Fig. 3. Loadmap of the PTD 420RX with heating powers $Q_1 = 0$ W and 20 W to the 1st stage while the load of the 2nd stage was varied between $Q_2 = 0$ W-5 W. The 2 curves show the 1st stage temperatures with respect to either the 2nd stage temperatures (black squares) or the intermediate stage temperatures (red circles), where no load was applied. The electric input power of the F-100 compressor was in the range of 11 to 13 kW. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

thermal stage (Fig. 1, right). The electrical heating as well as the temperature sensor are then located on the rectangular stage as indicated in Fig. 1., where the small black squares in the back of the regenerator stage are the electric heaters and the temperature sensor (Cernox, Lake Shore Cryotronics, Inc.) can be seen in front. The diameter of the regenerator tube is not changed and also the filling of the regenerator by rare earth materials remains completely unchanged. Due to the better thermal transport properties of copper compared to stainless steel the geometrical size of the regenerator heat exchanger leads to some thermal averaging across the HoCu2 part of the regenerator. The center of the regenerator heat exchanger is located appr. 65 mm away from the 2nd cooling stage, which is approximately 27% of the total length.

The final configuration of this modified PTD 420RX is schematically shown in Fig. 2. In addition to the rare earth materials and the three cooling stages, also stainless steel (SS) and Lead (Pb) are indicated as regenerator materials.

3. Results and discussion

Using the PTD 420RX configuration introduced in the previous section, the cooling capacity is subsequently discussed based on load maps systematically recorded by applying heat simultaneously to the 1st and 2nd stage as indicated in Fig. 3. The load maps in the diagram of Fig. 3 show the 2nd stage temperature T_2 (black squares) and the intermediate stage temperature T_3 (red circles), both plotted versus the 1st stage temperature T_1. The resulting load map closely resembles the data for the prototype PTC. In particular the important indicator of cooling power for the 2nd stage at 4.2 K was found to be 1.6 W. This comparison allows the conclusion that the influence of the additional stage on regular no load cooler operation, i.e. with $Q_3 = 0$ W, seems to be negligible.

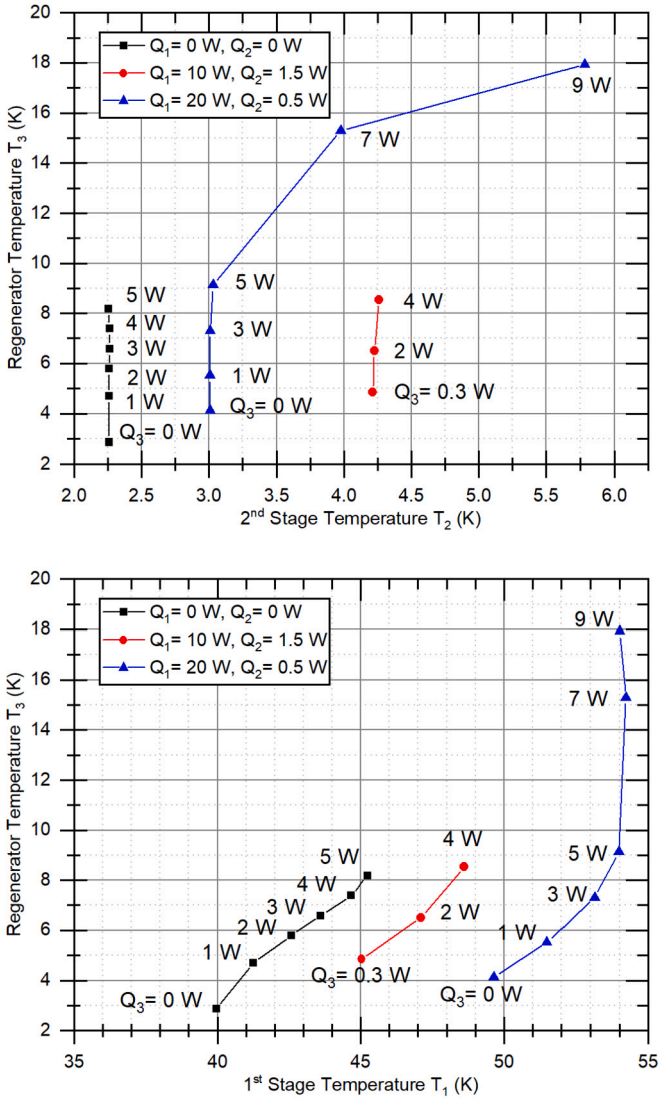


Fig. 4. Load-maps of the PTD 420RX obtained for systematic variation of the intermediate stage heating power Q_3 and different fixed loads of Q_1 and Q_2. Top: Regenerator temperature T_3 versus 2nd stage temperature T_2, Bottom: Regenerator temperature T_3 versus 1st stage temperature T_1.

However, to be truly advantageous an intermediate cooling stage requires to provide additional cooling power without influencing the performance of the 2nd stage. To analyze this question, three different curves are plotted in Fig. 4, where the intermediate stage heat load Q_3 was systematically varied while the 1st and 2nd stage heat loads Q_1 and Q_2 were kept constant in each case.

The curves of the top diagram of Fig. 4 allow to derive directly that the 2nd stage temperature remains mostly unaffected by regenerator heating for heat loads $Q_3 < 5$ W. On contrary, the curves plotted in the bottom diagram of Fig. 4 exhibit a direct shift to higher temperatures T_1 of the 1st stage once heat Q_3 is applied. However, in many applications the requirements for the 1st stage temperature are not as crucial and a moderate increase by 3-4 K is usually tolerable.

A Conclusion that can be drawn from the temperature response plotted in Fig. 4 is that applying additional heat load at the regenerator stage Q_3 rather shifts the thermodynamic balance with respect to the 1st stage while the 2nd stage appears to be thermally unaffected from this additional heating power source which coincides with previous findings [12].

In all scenarios covered by the plots displayed in Fig. 4 (top and bottom), an additional cooling power Q_3 of at least 4 W can be extracted at

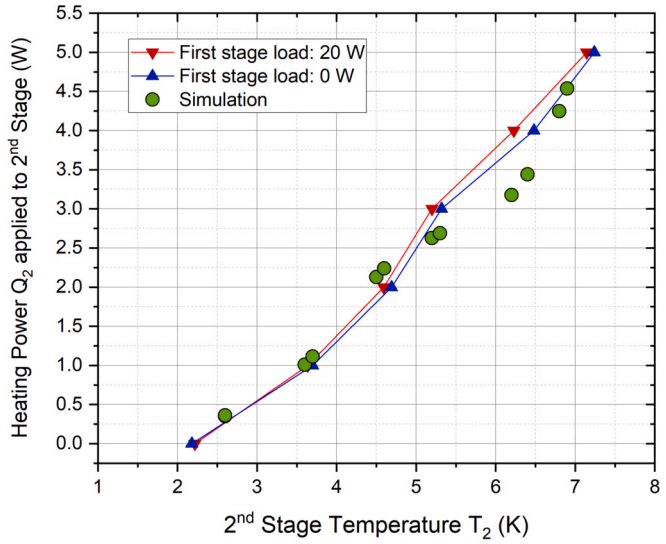


Fig. 5. Comparison between the experimentally measured cooling powers and REGEN simulations.

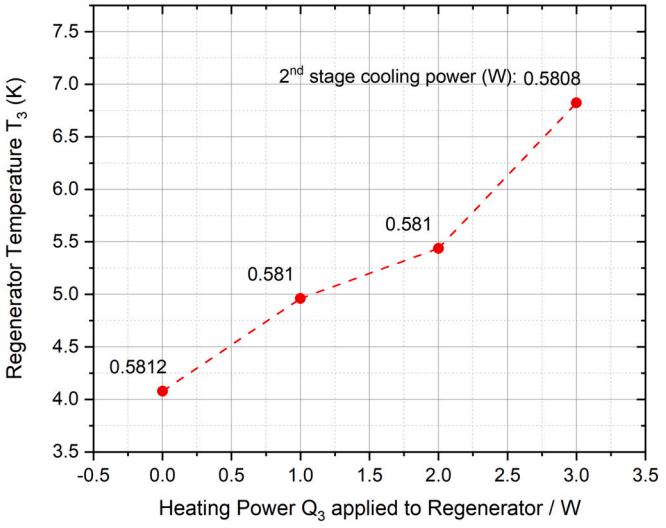


Fig. 6. Simulated temperature of the intermediate stage as a function of the applied heat load Q_3. The 1st and 2nd stage temperatures were fixed to 45 K and 3 K respectively during the simulations. The numbers adjacent to the data points refer to the cooling power of the 2nd stage and remain almost constant.

a temperature below 9 K which, for instance, allows to thermally anchor NbTi wires below their superconducting phase-transition-temperature.

The observed steep increase in temperature above appr. 7K of the intermediate heat exchanger can be anticipated based on theoretical derivations which consider enthalpy flows and temperature profiles across the regenerator [12]. Due to real gas effects and non-temperature-gradient enthalpy flow, the temperature profile at the cold end is initially expected to be flat. In this case, moderate heat injection from the intermediate heat exchanger does not alter the cooling power of the cold end and also the temperature of the intermediate heat exchanger only increases slightly. Instead, the temperature profile mostly changes closer to the warm end (see also Fig. 7).

However, if a certain threshold of intermediate heating power is exceeded, a transition starts where the initially flat profile at the cold end develops a gradient towards the cold end. In this case, the intermediate stage temperature T_3 increases faster, while the intermediate heat injection also directly reduces the cooling power at the cold end and T_2 increases as well.

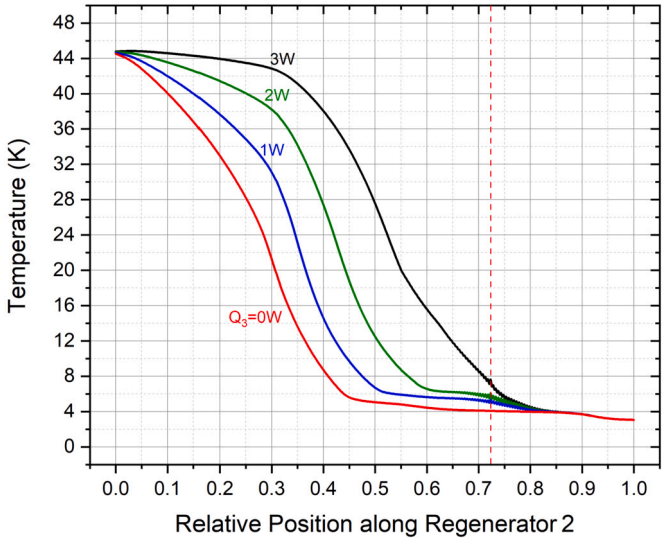


Fig. 7. Simulated temperature profiles along the 2nd stage regenerator calculated using REGEN with $T_1 = 45$ K and $T_2 = 3$ K. As indicated, the profiles are shown for four different values of Q_3 plotted against the relative length of the regenerator. The vertical dashed red line indicates the position of the heat source Q_3.

Interestingly, Fig. 3 shows that the temperature of the second stage can exceed the temperature of the intermediate stage for sufficiently large respective heating powers (P > 3W, T > 5.5K). In the temperature range analyzed in this work, such effects might also sensitively depend on changes of non-temperature-gradient power flow terms related to real gas effects and finite heat capacity. In addition also the specific material properties can play a role. In principle, this effect suggests, that the intermediate cooling stage is not only useful in combination with the 2nd stage but can be a sensible choice on its own for certain cooling scenarios.

To further corroborate these experimental cooling characteristics of the PTD 420RX cold head, numerical simulations using the Software REGEN3.3 were performed for comparison [15]. REGEN3.3 offers a well suited simulation environment allowing for the description of thermodynamic processes within a freely configurable regenerator [16]. In particular, it provides a range of functionalities of which especially the application of heat sources along the regenerator is relevant for this work.

As initial step, a simulation model for the PTD 420RX was generated, recreating its general cooling performance. Therefore, the 2nd stage regenerator was replicated in REGEN with both matching length and filling, while the temperature of the warm end was chosen according to the load maps. Within this configuration, the phase and the mass flow were adjusted until a good match between simulation and experiment was found for the cold end cooling power (Fig. 5).

This configuration served as model to calculate the additional cooling powers of a regenerator stage positioned according to the experiments. In these simulations, the cold and warm end temperatures were set to 45 K and 3 K respectively, while the additional regenerator heat load was varied.

The results are shown in Fig. 6 and confirm the general magnitude of heating power that can be extracted from the regenerator. Beyond that, Fig. 6 also shows that the 2nd stage cooling power is not influenced by Q_3 within the range of applied loads as can be concluded from the constant heat required to keep T_2 constant in the simulation. The latter is then further substantiated by Fig. 7, which shows the temperature profiles along the regenerator axis for different heat loads Q_3 of up to 3 W applied to the intermediate stage with trends similar to Ref. [17]. The temperature profile increase related to the additional heat load is mostly distributed to the warmer part of the regenerator, while temperatures

at the cold coincide. (Please note, that the 1st stage temperature was kept constant during the simulations, which does not fully reflect the experiments).

4. Conclusion

The comparison of experimental data with numerical simulations allows for a quantitave analysis of the potential that arises from complementary regenerator stages to provide additional cooling powers in two-stage pulse tube cold heads. Emanating from an initial cursory analysis, the optimum position along the regeneratior axis of such an additional stage is revealed. The results suggest a position at the very cold end of the 2nd stage regenerator but not too close to the 2nd cold stage. Based on these findings, a modified cold head was build with integrated stage from copper incorporated into the regenerator tube to ensure optimized thermal contact with the regenerator material and working fluid. Specification experiments with this cold head reproduce the initial cooling performance as well as allow for extraction of additional cooling power of up to 5 W at temperatures below 9 K at the new stage. Most importantly, this can be done without affecting the main cooling power of the 2nd stage.

Already in the current configuration, the cold head is well suited to enable additional features to increase cryostats efficiency by e.g. thermally anchoring superconducting wires with transition temperatures around 9 K (e.g. NbTi).

Recently, also more and more multichannel systems were introduced, that can benefit from such an additional cooling. The possible applications range from precooling multiple wires for Squid and/or superconducting nanowire single-photon detector (SNSPD) arrays, optical fibers or microwave guides, or high current bus bars for magnet solenoids. The thinkable range of applications also extends to additional radiation shielding - equipped with optical windows. Made from NbTi or lead, a hermetic housing could also efficiently shield magnetic fields.

From the methodological point of view, this first approach allows for further technical optimizations. Depending on different applications and according heat load scenarios, highly adapted geometries of the incorporated regenerator stage might be useful. New 3D print technologies from copper widen the range how such integration can be shaped.

In fact, the thermal link to the regenerator filling and working fluid inside the cold heads tube can be investigated to enhance the efficiency of extracting this extra cooling power. In this context, regenerator fillings of mesh, grains or spheres, and powder can come into play for different materials like, pure metals, alloys, ceramics or plastics, or combinations thereof. The interplay of thermal conductivity and heat capacity with respect to the machinable geometry is a major point to be addressed for future improvements.

Finally, the incorporation of additional stages is not limited to only one stage as demonstrated by Ravex et al. for two stages [7]. Multiple stages propose ultimately increased flexibility for the adjustment of optimal integration of closed cycle coolers even in complex cryostat designs. Such multiple stages could be used separately or simultaneously depending on the exact application [13].

However, especially for this final idea new prototypes together with further simulations by REGEN and Sage would be required to explore how simultaneously applied load to different regenerator stages influences the temperature distributions and especially the main cooling stages.

CRediT authorship contribution statement

Jens Falter: Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Jack Schmidt:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Xaver Herrmann:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Bernd Schmidt:** Writing – review & editing, Validation, Formal analysis. **Dirk Dietzel:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **André Schirmeisen:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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